

MHF4U Advanced Functions Final Exam

Name: _____

Date: _____

Please read all the following instructions carefully before you start your exam.

- This is not an open book exam, other than a standard calculator or scrap paper, aids such as cheat sheets, a programmable calculator, electronic dictionaries, are not allowed.
- You are responsible to bring your own calculator. Sharing a calculator with someone else is strictly prohibited.
- Graphing calculators are NOT allowed for this exam.
- Please show all your work on the space that is provided in order to get full marks.

GOOD LUCK!

Knowledge	Application	Thinking	Communication
/29	/17	/19	/10

9. Given that $\sin \theta = \frac{3}{8}$ and θ is obtuse, what is the value of $\tan 2\theta$?
- a. $-\frac{3\sqrt{55}}{23}$ b. $\frac{3\sqrt{55}}{46}$ c. $-\frac{3\sqrt{55}}{55}$ d. $\frac{3\sqrt{55}}{55}$
10. Which expression is equivalent to $\cos x$?
- a. $\frac{2\cos^2(\frac{1}{2}x)-1}{\cos^2(\frac{1}{2}x)}$ b. $2\cos^3(2x) - 1$ c. $\frac{2-\sec^2(\frac{1}{2}x)}{\sec^2(\frac{1}{2}x)}$ d. $1 - 2\sin^2(2x)$
11. Which set of values, in radians, is the solution of $5 + 7\sin x = 0$, where $-\pi \leq \theta \leq \pi$?
- a. -0.8 b. $-0.8, -2.35$ c. $0.8, 2.35$ d. $-0.8, 0.8$
12. What is the exponential form of $y = \log_7 x$?
- a. $x = \log_7 y$ b. $x = 7^y$ c. $y = 7^x$ d. $y = x^7$
13. The function $f(x) = \log_{10} x$ is reflected in the x-axis, stretched horizontally by a factor of 3, and translated 2 units up. Which of these is the resulted transformed function?
- a. $g(x) = -\log_{10}(3x) - 2$ b. $g(x) = \log_{10}\left(-\frac{1}{3}x\right) + 2$
- c. $g(x) = -\log_{10}\left(\frac{1}{3}(x-2)\right) + 2$ d. $g(x) = -\log_{10}\left(\frac{1}{3}x\right) + 2$
14. A transformation that takes the graph of $f(x) = \log_5 x$ to that of $g(x) = \log_5 25x$ is...
- a. Horizontally compressed by a factor of $1/25$
- b. Vertically compressed by a factor of $1/25$
- c. Vertically stretched by a factor of 25
- d. Horizontally stretched by a factor of 25
15. Solve $x = \log_3 27\sqrt{3}$
- a. $2\frac{1}{2}$ b. $3\frac{1}{2}$ c. $3\frac{1}{3}$ d. $9\frac{1}{9}$
16. If $f(x) = \log x$ and $g(x) = \frac{1}{x-3}$, which set is the domain of $f - g$?
- a. $\{x \in \mathbb{R} \mid x > 3\}$ b. $\{x \in \mathbb{R} \mid x > 0, x \neq 3\}$
- c. $\{x \in \mathbb{R} \mid x > 0, x \neq -3\}$ d. $\{x \in \mathbb{R} \mid x < 3\}$
17. If $f(x) = \sqrt{3-x}$ and $g(x) = 3x^2$, what is the domain $f \circ g$?
- a. $\{x \in \mathbb{R} \mid -3 \leq x \leq 3\}$ b. $\{x \in \mathbb{R} \mid x \leq 3\}$
- c. $\{x \in \mathbb{R} \mid -1 \leq x \leq 1\}$ d. $\{x \in \mathbb{R} \mid x \geq 0\}$

18. Solve the following: $\log_a(x - 3) + \log_a(x - 2) = \log_a(5x - 15)$ (3 marks)

19. Find the domain of the following equations (4 marks)

a. $y = \frac{1}{\sqrt{x^2 - 9}}$

b. $y = 2x\sqrt{x - 8} + \frac{1}{x - 12}$

20. Given $f(x) = ax^2 + 3$ and $g(x) = bx - 1$, the graph of the product $f \cdot g$ passes through the points $(-1, -3)$ and $(1, 9)$ (5 marks)

a. What are the values of a and b ?

b. Graph the combined function $f \cdot g$

APPLICATION

21. A gymnasium has a circular running track around the mezzanine with a radius of 20 m. A runner ran along the arc of the track for 60 m. What is the sector angle, in radians? (2 marks)

22. Determine an exact value for $\sin \frac{3\pi}{4} - \tan \frac{5\pi}{6}$ (3 marks)

23. Angle a is located in the 2nd quadrant such that $\sin a = \frac{12}{13}$. Angle b is located in the 3rd quadrant such that $\cos b = -\frac{4}{5}$. Determine an exact value for $\sin(a + b)$ (4 marks)

24. Prove the following identities (8 marks)

a. $\frac{\sin 2x}{\sec x} = \frac{2 \cos^2 x}{\csc x}$

b. $\sin(x + y) \cos(x - y) = \frac{\sin x}{\sec x} + \frac{\cos y}{\csc y}$

THINKING

25. The equation $\cos 2\theta + d \cos \theta + e = 0$ has solutions $\theta = 0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$ in the interval $0 \leq \theta \leq 2\pi$. What are the values of d and e ? (4 marks)

26. Solve for x (9 marks)

a. $5^{4-x} = \frac{1}{5}$

b. $3^{2x} = 7$

c. $7 + 2^{x+4} = 12$

27. A wound, initially with an area of 80 cm^2 , heals according to the formula $A(t) = 80(10^{-0.023t})$, where $A(t)$ is the area of the wound in square centimetres after t days of healing. In how many days will 75% of the wound be healed? (3 marks)

28. Given $h(x) = |x^2 - 1|^3$, find 2 functions, f and g , such that $h = f \circ g$ (3 marks)

COMMUNICATION

29. Which of the following statements are true? Show your work (6 marks)

a. The sum of two odd functions is an odd function

b. The product of an even function and an odd function is an odd function

c. The quotient of two odd functions is an even function

30. Describe how to graph of $g(x) = \frac{3}{(x-5)^2} - 8$ can be obtained from the graph of $f(x) = \frac{1}{x}$, graph both $f(x)$ and $g(x)$ on the same axis. Give detailed graphical analysis of $g(x)$ (4 marks)